

AW
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Name: _____

Class: 12MT2__ or 12MTX__

Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2009 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS

*Time allowed - 3 HOURS
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 10.

*****Each page must show your name and your class. *****

Question 1 (12 marks)**Marks**

- (a) Expand and simplify $8x^2 + 2xy - 2x(4x + y)$. **2**
- (b) Completely factorise $5x^2 - 20$ **2**
- (c) Find the value of $\tan \frac{8\pi}{7}$ correct to three significant figures. **2**
- (d) Find the values of x for which $|2 - 3x| > 11$. **2**
- (e) Express $3\sqrt{32} - \sqrt{128}$ in simplest surd form. **2**
- (f) If $k = \frac{1}{3}m(v^2 - u^2)$ find the value of m when $k = 1224$, $v = 14.2$ and $u = 7.4$. **2**

Question 2 (12 marks) (START A NEW PAGE)

Marks

(a) Differentiate with respect to x .

(i) $(2x^2 + 1)^8$

2

(ii) $x^2 e^{2x}$

2

(iii) $\frac{x}{\ln x}$

2

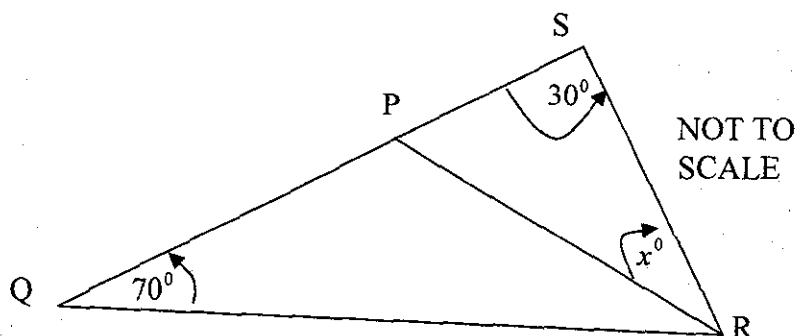
(b) (i) Find $\int (2x+1)^5 dx$

2

(ii) Evaluate $\int_0^{\frac{\pi}{8}} 4\cos 4x dx$

2

(c)



Given $PQ = PR$, $\angle PQR = 70^\circ$, $\angle PSR = 30^\circ$ and $\angle PRS = x^\circ$,

2

find the value of x giving reasons.

Question 3 (12 marks) (START A NEW PAGE)

Marks

(a) Given the points A(1,2), B(3,1), C(-1,4)

(i) Find the equation of the line BC.

1

(ii) Find the perpendicular distance from point A to the line BC.

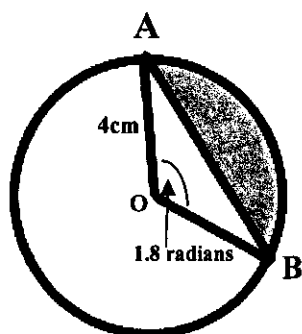
2

(iii) Hence, or otherwise, find the area of $\triangle ABC$.

2

(b) A chord AB subtends an angle of 1.8 radians at the centre of a circle of radius 4 cm. Find the area of the minor segment cut off by the chord AB. (Give your answer correct to 1 decimal place.)

2



(c) For the function $y = 2\sqrt{9 - x^2}$

state the

(i) domain and

2

(ii) range.

1

(d) Find the radius and centre of the circle with equation

2

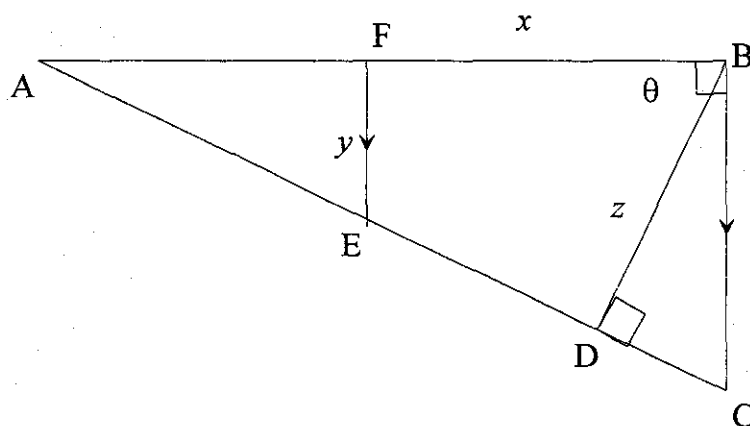
$$x^2 - 4x + y^2 + 6y + 12 = 0.$$

Question 4 (12 Marks)

(START A NEW PAGE)

Marks

- (a) The right triangle ABC is shown below. $BC \parallel FE$, $BD \perp AC$, $\angle FBD = \theta$, $BF = x$, $EF = y$ and $BD = z$.



Prove

(i) $\triangle BDA \parallel \triangle EFA$

2

Show that

(ii) $\angle FEA = \theta$

1

(iii) $AB = x + y \tan \theta$

1

(iv) $z = x \cos \theta + y \sin \theta$

1

- (b) Find the equation of the normal to the curve $y = \sqrt{x}$ at the point (4,2).

2

- (c) A ship sails from port A, 60 nautical miles due west, to a port B. It then proceeds a distance of 50 nautical miles on a bearing of 210° to a port C.

- (i) Draw a diagram to illustrate the information given.

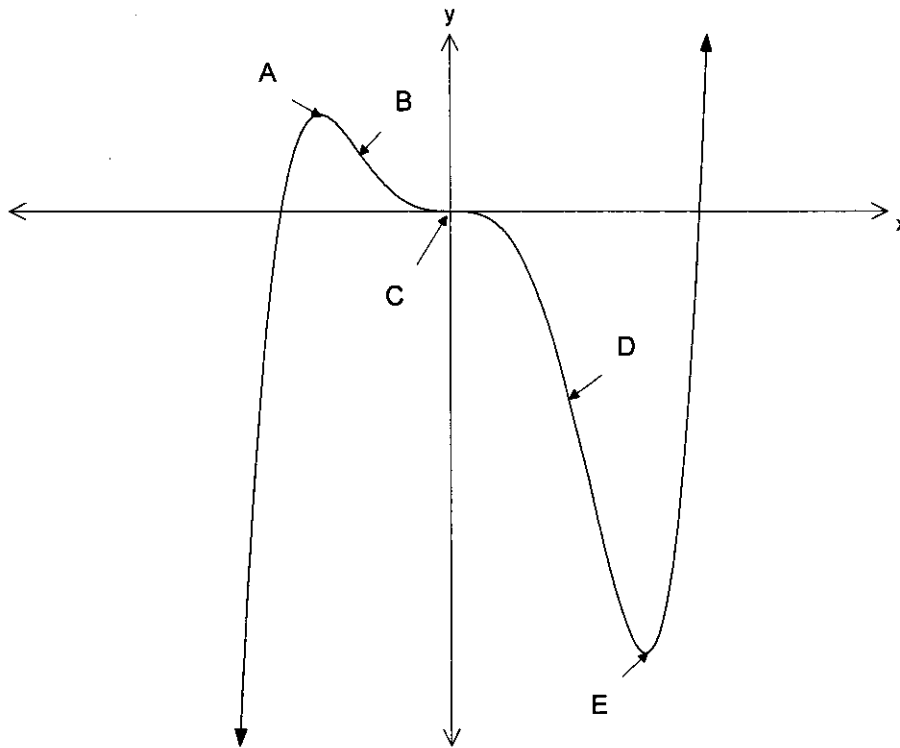
1

- (ii) Find the distance (nearest nautical mile) and bearing of C from A.

4

Question 5 (12 Marks)**(START A NEW PAGE)****Marks**

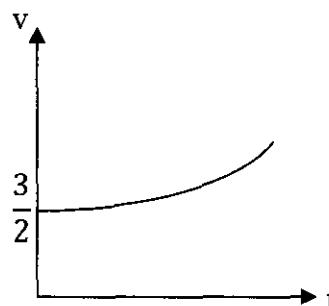
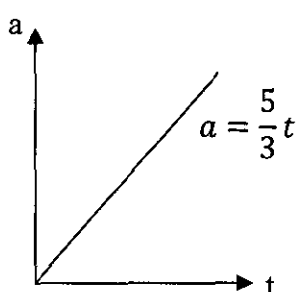
- (a) The graph of the curve $y = f(x)$ is drawn below.



- | | | |
|-------|-------------------------------|---|
| (i) | Name the points of inflexion. | 1 |
| (ii) | Name the stationary points. | 1 |
| (iii) | Sketch the gradient function. | 1 |
- (b) The rate at which Carbon Dioxide will be produced when conducting an experiment is given by $\frac{dV}{dt} = \frac{1}{100}(30t - t^2)$ where $V \text{ cm}^3$ is the volume of gas produced after t minutes.
- | | | |
|------|--|---|
| (i) | At what rate is the gas being produced 15 minutes after the experiment begins? | 1 |
| (ii) | How much Carbon Dioxide has been produced during these 15 minutes? | 2 |

Question 5 is continued on page 6.....

- (c) The graphs below describe the flight of a sugar glider after it jumps from a tall tree. Assume that the displacement at time $t = 0$ is 0 metres.



- (i) Show that the velocity of the sugar glider for $t \geq 0$ is given by $v = \frac{1}{6}(5t^2 + 9)$. 2
- (ii) Determine the expression for the displacement (x) for $t \geq 0$ and hence calculate the displacement of the sugar glider at $t = 2$. 2
- (d) The student population “ C ” at Cherrybrook Technology High School in the first 6 months of 2009 was increasing at a decreasing rate.
What does this tell you about

$$\frac{dC}{dt} \text{ and } \frac{d^2C}{dt^2} \quad .$$

2

Question 6 (12 Marks)**(START A NEW PAGE)****Marks**

(a) For the curve $y = 2x^3 - 9x^2 + 12x$,

(i) Show that $\frac{dy}{dx} = 6(x-1)(x-2)$

1

(ii) Find the coordinates of the stationary points and determine their nature.

3

(iii) Graph the function showing clearly the stationary points.

2

(iv) Where is $y=f(x)$ decreasing?

1

(b) Consider the parabola $y^2 = 8(x+2)$

(i) Find the coordinates of the vertex.

1

(ii) Find the coordinates of the focus.

1

(iii) Find the equation of the directrix.

1

(c) Find the value(s) of m for which the equation $4x^2 - mx + 9 = 0$ has exactly one real root.

2

- (a) The population, P , of a certain town grows at a rate proportional to the population, ie. $\frac{dP}{dt} = kP$.

If the population grows from 20 000 to 25 000 in two years, find:

- (i) the value of k (the growth constant) correct to 4 significant figures. 2
- (ii) the population of the town, to the nearest hundred, after a further 8 years. 1
- (iii) Calculate the rate of change of the population at this time (ie after these further 8 years) correct to the nearest hundred. 1

- (b) A particle moves in a straight line so that its displacement, x metres from a fixed point O on a line, is given by

$$x = t + \frac{16}{t+1}$$

where t is measured in seconds.

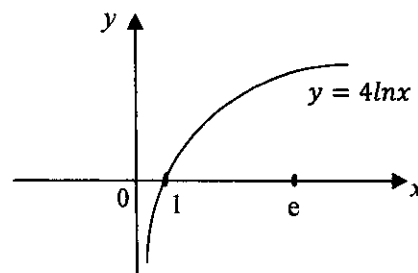
- (i) Find the particle's initial position 1
- (ii) Find expressions for the velocity and acceleration of the particle in terms of t . 2
- (iii) Find when and where the particle is at rest. 2
- (iv) Find the limiting velocity of the particle. 1
- (v) Find the total distance travelled by the particle in the first 4 seconds. 2

Question 8 (12 Marks)**(START A NEW PAGE)****Marks**

- (a) The gradient of a curve is given by $\frac{dy}{dx} = \frac{2x}{x^2 + e}$. 2
The curve passes through the point (0,2).

What is the equation of the curve?

- (b) The arc of the curve $y = 4 \ln x$, between $x = 1$ and $x = e$, is rotated about the y-axis.



- (i) Show that the volume formed is given by $V = \pi \int_0^4 e^{0.5y} dy$ 2
- (ii) Hence, find the exact volume. 2
- (c) (i) Is the series $\log 3 + \log 9 + \log 27 + \dots$ arithmetic or geometric? 2
Give reasons for your answer.
- (ii) Find the sum of the first 10 terms of the series in exact form. 2
- (d) Use the trapezoidal rule with four function values. to find an approximate value of 2
the area under the curve $y = 3^x$, bounded by the x axis, $x = 1$ and $x = 4$

Question 9 (12 marks) (START A NEW PAGE)

Marks

(a) Solve $2\cos x + \sqrt{3} = 0$ for $-\pi \leq x \leq \pi$ 2

(b) (i) Differentiate $e^{\tan x}$ 1

(ii) Hence or otherwise, evaluate $\int_0^{\frac{\pi}{4}} 3e^{\tan x} \sec^2 x \, dx$ 2

(c) The minute hand on a clock face is 12 centimetres long.

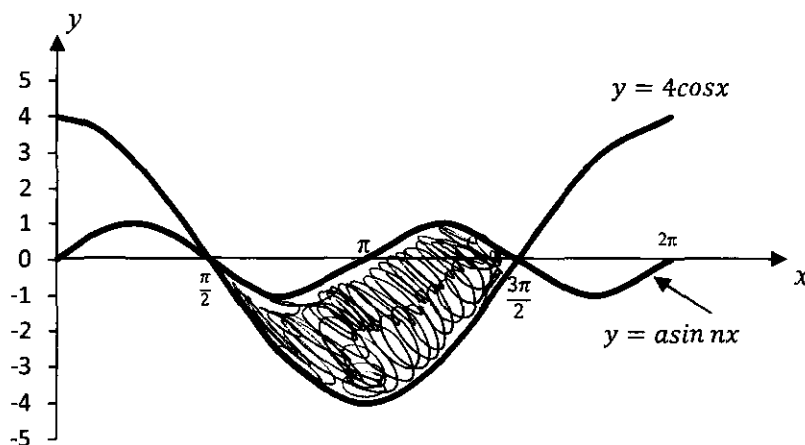
In 40 minutes,

(i) through what angle does the hand move (in radians)? 1

(ii) how far does the tip of the hand move? 1

(iii) what area does the hand sweep through in this time? 1

(d)



The graph above shows $y = 4\cos x$ and $y = a\sin nx$.

(i) Find the values of a and n . 1

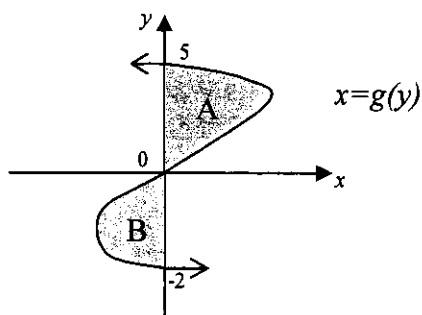
(ii) Find the area of the shaded region in the diagram. 3

Question 10 (12 marks) (START A NEW PAGE)

Marks

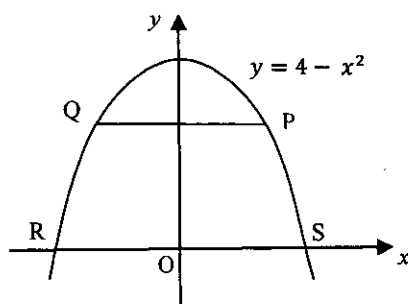
- (a) The area A is equal to 7 square units and area B is equal to 9 square units.

1



Evaluate $\int_{-2}^5 g(y) dy$.

- (b)



Not to scale

The parabola $y = 4 - x^2$ cuts the x -axis at R and S. The point $P(x,y)$ lies on the parabola in the first quadrant. Q also lies on the parabola such that PQ is parallel to the x -axis.

- (i) Write down the co-ordinates of R and S.

1

- (ii) Show that the area of trapezium PQRS is given by:

2

$$A = (2 + x)(4 - x^2)$$

- (iii) Hence, find the value of x which gives a maximum value of A , justifying your answer.

3

Question 10 is continued on page 12.....

- (c) A small oil company has 50 000 barrels of oil in stock. In its first month of operation, the company increased its stock by 5%, and then sold 1500 barrels.

Each subsequent month, the company continued to increase its previous month's stock by 5%, and then also increased the number of barrels of oil sold by 3%.

- (i) Show that the number of barrels of oil in stock after 3 months can be expressed as:

3

$$B_3 = 50000 \times 1.05^3 - 1500 \times (1.05^2 + 1.05 \times 1.03 + 1.03^2)$$

- (ii) Show that the number of barrels of oil in stock after n months can be expressed as:

2

$$B_n = 50000 \times 1.05^n - 75000 \times 1.05^n \left(1 - \left(\frac{1.03}{1.05} \right)^n \right)$$

End of Examination.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x$, $x > 0$

SOLUTIONSQuestion 1

$$\begin{aligned}
 (a) \quad & 8x^2 + 2xy - 2x(4x + y) \\
 & = 8x^2 + 2xy - 8x^2 - 2xy \\
 & = 0
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & 5x^2 - 20 = 5(x^2 - 4) \\
 & = 5(x - 2)(x + 2)
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & \tan \frac{8\pi}{7} = 0.4815746, \dots \\
 & = 0.482
 \end{aligned}$$

$$(d) \quad |2 - 3x| > 11$$

$$2 - 3x < -11 \quad \text{or} \quad 2 - 3x > 11$$

$$-3x < -13 \quad \text{or} \quad -3x > 9$$

$$x > \frac{13}{3} \quad \text{or} \quad x < -3$$

$$\begin{aligned}
 (e) \quad & 3\sqrt{32} - \sqrt{128} = 12\sqrt{2} - 8\sqrt{2} \\
 & = 4\sqrt{2}
 \end{aligned}$$

$$(f) \quad K = \frac{1}{3}m(v^2 - u^2)$$

$$1224 = \frac{1}{3}m(14.2^2 - 7.4^2)$$

$$1224 = 48.96m$$

$$m = \frac{1224}{48.96}$$

$$m = 25$$

Question 2

$$(a) (i) = 8(2x^2+1)^7 \times 4x \checkmark$$

$$= 32x(2x^2+1)^7 \checkmark$$

award (2)
for answer only

$$(ii) x^2 \times 2e^{2x} + e^{2x} \times 2x \checkmark$$

$$= 2x^2 e^{2x} + 2x e^{2x} \checkmark$$

$$\text{or } = 2x e^{2x} (x+1) \checkmark$$

award (2)
answer only

$$(iii) \ln x \times 1 - x \times \frac{1}{x}$$

$$\frac{(\ln x)^2}{(\ln x)^2} \checkmark$$

do not accept $\ln x^2$
(if correct with $\ln x^2$ give (1))

award (2)
answer only

$$(b) (i) \int (2x+1)^5 dx = \frac{(2x+1)^6}{6 \times 2} + C \checkmark$$

ignore missing + C

$$= \frac{1}{12} (2x+1)^6 + C \checkmark$$

award (2)
answer only

$$(ii) \int_0^{\pi/8} 4 \cos 4x dx = \left[\sin 4x \right]_0^{\pi/8} \checkmark$$

$$= \sin \frac{\pi}{2} - \sin 0$$

$$= 1 \checkmark$$

(c) PQ = QR

$$\therefore \angle PQR = \angle PRQ = 70^\circ \text{ (In any } \Delta, \text{ angles opposite equal sides are equal)} \checkmark$$

ΔPQR IS ISOSCELES (PQ = QR)
 $\therefore \angle PQR = \angle PRQ = 70^\circ$ (equal base angles of isosceles Δ)

In ΔQSR

$$70 + 70 + x + 30 = 180 \text{ (angle sum } \Delta) \checkmark$$

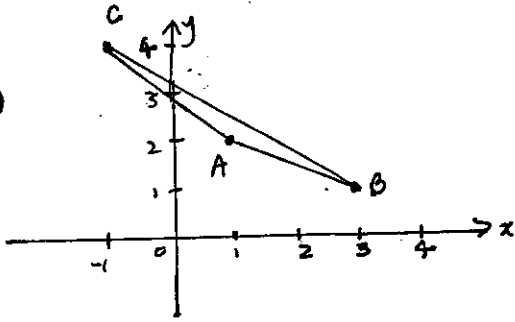
$$\therefore x = 10$$

for 2nd mark
needs to see some words + words angle sum Δ then $x=10$

for 1st mark
must all do this way
need to see
isos because of = sides
 $\angle PRQ = 70^\circ$ (equal base angles)

QUESTION 3

(a) (i)



$$\text{slope } BC = -\frac{3}{4}$$

equation BC

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{3}{4}(x + 1)$$

$$4y - 16 = -3x - 3$$

$$3x + 4y - 13 = 0 \quad \checkmark \quad (\text{or equivalent expression})$$

$$(ii) \quad p = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$= \frac{|3 \times 1 + 4 \times 2 - 13|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{|3 + 8 - 13|}{\sqrt{25}}$$

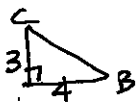
$$= \left| \frac{-2}{5} \right|$$

$$= \frac{2}{5} \text{ units}$$

x_1, y_1
 $A(1, 2)$

correct
but no
absolute
sign
Ans ①

(iii) Find BC using Pythagoras' Theorem or distance Formula



$$BC = 5 \text{ units}$$

(Pythagorean Triad {3, 4, 5})

$$\text{Area} = \frac{1}{2} \times \frac{3}{5} \times 5$$

$$= 1 \text{ sq. unit}$$

$$(b) \text{ Area} = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 4^2 \times (1.8 - \sin 1.8)$$

$$= 6.60921 \dots$$

$$= 6.6 \text{ cm}^2 \text{ (1 dec pl)}$$

$$(c) (i) 9 - x^2 \geq 0$$

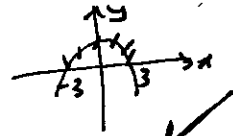
$$(3 - x)(3 + x) \geq 0$$

$$-3 \leq x \leq 3$$

(or $x \geq -3$ and $x \leq 3$)

$$(ii) 0 \leq y \leq 6$$

(or $y \geq 0$ and $y \leq 6$)



$$(d) x^2 - 4x + y^2 + 6y + 12 = 0$$

$$x^2 - 4x + 4 + y^2 + 6y + 9 = -12 + 4 + 9$$

$$(x - 2)^2 + (y + 3)^2 = 1$$

$\therefore \text{radius} = 1 \text{ unit}$
 $\text{centre} = (2, -3)$

QUESTION 4

(a) (i) In $\triangle BDA$, EFA
 $\angle BDA = 90^\circ$ (given $BD \perp AC$)
 $\angle AFE = \angle ABC$ (corresponding angles, $FE \parallel BC$)
 $= 90^\circ$

- $\angle AFE = \angle BDA = 90^\circ$
- $\angle BAD$ is common
- $\therefore \triangle BDA \parallel \triangle EFA$ (equiangular)

(ii) $\angle DBA = \angle FEA = \theta$
 (corresponding angles of similar $\triangle$'s)

(iii) In $\triangle AFE$
 $\tan \theta = \frac{AF}{y}$
 $AF = y \tan \theta$
 $\therefore AB = y \tan \theta + x$

(iv) In $\triangle ABD$
 $\cos \theta = \frac{z}{AB}$
 $= \frac{z}{y \tan \theta + x}$

$$z = y \tan \theta \cos \theta + x \cos \theta$$

$$= y \frac{\sin \theta}{\cos \theta} \cos \theta + x \cos \theta$$

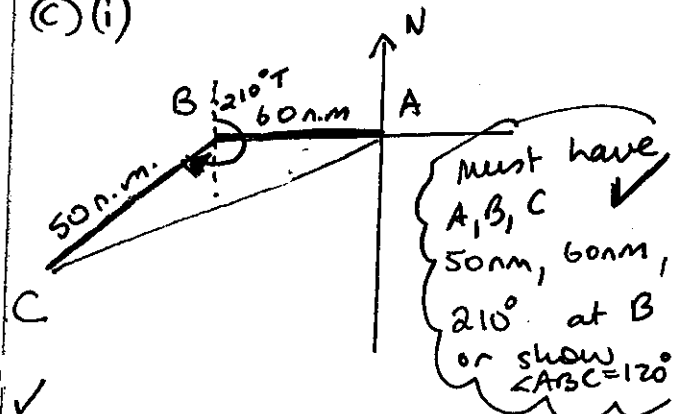
$$= y \sin \theta + x \cos \theta$$

(b) $\frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$
 $= \frac{1}{2\sqrt{x}}$

when $x = 4$
 slope of tangent $= \frac{1}{2\sqrt{4}} = \frac{1}{4}$
 $\therefore$ slope of normal $= -4$ ($m_1 m_2 = -1$)

$x = 4, y = 2$
 $y - 2 = -4(x - 4)$
 $y - 2 = -4x + 16$
 $4x + y - 18 = 0$
 or
 $y = -4x + 18$
 or
 (equivalent eqn)

(c) (i)



(d) $AC^2 = 60^2 + 50^2 - 2 \times 60 \times 50 \times \cos 120^\circ$

$AC = 95.3939 \dots$
 $AC = 95 \text{ n.m. (nearest nm)}$

$\frac{\sin \angle BAC}{50} = \frac{\sin 120^\circ}{AC}$

$\sin \angle BAC = \frac{50 \sin 120^\circ}{AC}$

$\angle BAC = 27^\circ$

Bearing of C from A is 243°T or 563°W

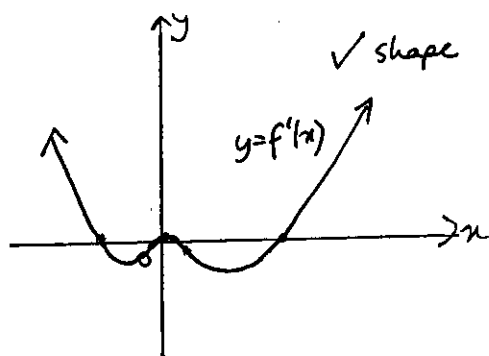
(answer using 95nm rounded is same $\rightarrow$ Accept!)

QUESTION 5

(a) (i) B, C and D ✓

(ii) A, C and E. ✓

(iii)



$$(b) \frac{dv}{dt} = \frac{1}{100} (30t - t^2)$$

$$(i) \frac{dv}{dt} = \frac{1}{100} (30 \times 15 - 15^2)$$

$$= 2\frac{1}{4}$$

$$2\frac{1}{4} \text{ cm}^3/\text{min}$$

$$(ii) \int_0^{15} \frac{1}{100} (30t - t^2) dt$$

$$= \frac{1}{100} \left[15t^2 - \frac{t^3}{3} \right]_0^{15}$$

$$= \frac{1}{100} \left[(15 \times 15^2 - \frac{15^3}{3}) - (0) \right]$$

$$= \frac{1}{100} \left[3375 - \frac{3375}{3} \right]$$

$$= 22.5 \checkmark$$

$\therefore 22.5 \text{ cm}^3$ is produced in 15 minutes

$$\text{OR } v = \int \frac{1}{100} (30t - t^2) dt$$

$$= \frac{1}{100} \left[15t^2 - \frac{t^3}{3} \right] + c \quad (\checkmark)$$

$$t=0 \quad v=0$$

$$0 = \frac{1}{100} [0 - 0] + c$$

$$c=0$$

$$\therefore v = \frac{1}{100} \left[15t^2 - \frac{t^3}{3} \right]$$

$$t=15 \quad v = \frac{1}{100} \left[15 \times 15^2 - \frac{15^3}{3} \right]$$

$$= 22.5 \text{ cm}^3 \quad (\checkmark)$$

alternate method and corresponding mark allocation

$$(c) (i) a = \frac{5}{3}t$$

$$v = \frac{5}{6}t^2 + c$$

$$t=0, v = \frac{3}{2} \text{ (diagram)}$$

$$t=0 \quad \frac{3}{2} = \frac{5}{6} \times 0^2 + c$$

$$c = \frac{3}{2}$$

$$v = \frac{5}{6}t^2 + \frac{3}{2} \quad \checkmark$$

$$v = \frac{1}{6}(5t^2 + 9)$$

(ii)

$$x = \frac{5}{18}t^3 + \frac{3}{2}t + c$$

$$t=0, x=0$$

$$\therefore c=0$$

$$x = \frac{5}{18}t^3 + \frac{3}{2}t$$

$$\text{when } t=2$$

$$x = \frac{5}{18}(2)^3 + \frac{3}{2}(2)$$

$$= 5\frac{2}{9}$$

$\therefore$ The displacement is $5\frac{2}{9} \text{ m}$ (in the positive direction) ✓

(d)

$$\frac{dc}{dt} > 0 \quad \checkmark$$

$$\frac{d^2c}{dt^2} < 0 \quad \checkmark$$

QUESTIONS 6

(a) (i) $y = 2x^3 - 9x^2 + 12x$
 $\frac{dy}{dx} = 6x^2 - 18x + 12$ ✓
 $= 6(x^2 - 3x + 2)$
 $= 6(x-2)(x-1)$

(ii) $\frac{dy}{dx} = 0$ for stationary points

$6(x-2)(x-1) = 0$

(2nd derivative method) $x = 1$ or $x = 2$ ✓

$\frac{d^2y}{dx^2} = 12x - 18$

$x = 1$

$\frac{d^2y}{dx^2} = 12 \times 1 - 18$

$= -6$

< 0

∴ relative maximum turning point at (1, 5)

$x = 2$

$\frac{d^2y}{dx^2} = 12 \times 2 - 18$

$= 6$

> 0

∴ relative minimum turning point at (2, 4)

or (1st derivative test)

x	0	1	1.5	2	3
f(x)	12	0	-1.5	0	12
slope	/	-	\	-	/

note they need to be careful testing to the right of $x=1$ and to the left of $x=2$

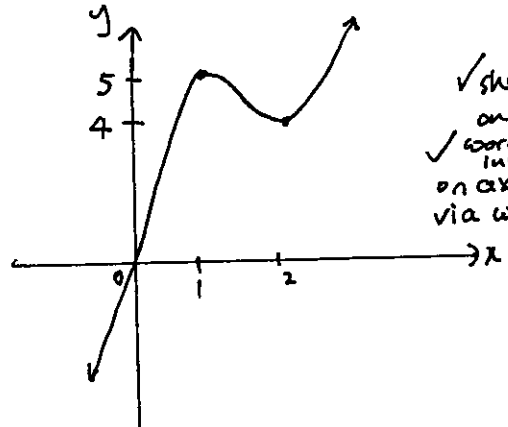
∴ relative maximum turning point at (1, 5)
 relative minimum turning point at (2, 4)

1 mark for correct test and a pair of coordinates (each time)

or
 1 mark for 2 correct tests (i.e. mistakes only in y-values)

but not 1 mark for 2 y values

(iii)



✓ shape and coordinates indicated on axes or via wordnotes

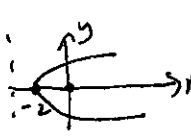
(iv) $1 < x < 2$ ✓

(b) (i) (-2, 0) ✓

(ii) $8 = 4a$
 $a = 2$

Focus (0, 0)

(iii) $x = -4$ ✓



(c) $\Delta = 0$

$m^2 - 4 \times 4 \times q = 0$ ✓

$m^2 = 144$

$m = \pm 12$

✓ must have both

QUESTION 7

$$\begin{aligned} \text{(11)} \quad \frac{dp}{dt} &= K P \\ &= \frac{1}{2} \ln\left(\frac{5}{4}\right) \times 61035.156... \checkmark \\ &= 6809.80076.... \end{aligned}$$

$$\begin{aligned} \text{or using (11) answer} \\ &= \frac{1}{2} \ln\left(\frac{5}{4}\right) \times 61000 \quad (\checkmark) \\ &= 6805.878.... \end{aligned}$$

$$\begin{aligned} \text{or using (11) answer + rounded } K \\ &= 0.1116 \times 61000 (\checkmark) \\ &= 6807.6 \end{aligned}$$

Answer for various "K"
Answer is correct to nearest hundred so $\rightarrow$ 6800 people/year.

Alternate variables

Students might also do (accept bald answer)
 $P = 20000e^{Kt}$
 $\frac{dp}{dt} = 20000Ke^{Kt}$ (accept with a sub for K)
 $= \dots$ (answers same as above)

$$\text{(b)} \quad x = t + \frac{16}{t+1}$$

$$\begin{aligned} \text{(i)} \quad t &= 0 \\ x &= 0 + \frac{16}{0+1} \\ x &= 16 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad v &= 1 - 16(t+1)^{-2} \checkmark \\ &= 1 - \frac{16}{(t+1)^2} \end{aligned}$$

$$\begin{aligned} a &= 32(t+1)^{-3} \checkmark \\ &= \frac{32}{(t+1)^3} \end{aligned}$$

$$\text{(iii)} \quad 0 = 1 - \frac{16}{(t+1)^2}$$

$$\begin{aligned} \frac{16}{(t+1)^2} &= 1 \\ (t+1)^2 &= 16 \end{aligned}$$

$$\begin{aligned} t+1 &= \pm 4 \\ t &= 3 \text{ or } -5 \end{aligned}$$

$$t \geq 0 \quad \therefore t = 3 \quad \checkmark$$

(a)

$$\frac{dp}{dt} = KP$$

$$\text{(i)} \quad P = 20000e^{Kt}$$

$$25000 = 20000e^{2K}$$

$$e^{2K} = \frac{25}{20}$$

$$= \frac{5}{4}$$

$$2K = \log_e\left(\frac{5}{4}\right)$$

$$K = \frac{1}{2} \log_e\left(\frac{5}{4}\right)$$

$$= 0.111571....$$

$$= 0.1116 \quad (\text{4 sig. fig})$$

$$\text{(ii)} \quad P = 20000e^{Kt}$$

$$t = 10, \quad K = \frac{1}{2} \log_e\left(\frac{5}{4}\right)$$

$$= \ln \frac{5}{4}$$

$$P = 20000e^{5 \ln \frac{5}{4}}$$

$$= 61035.156....$$

$\therefore$ Population is 61000 (correct to nearest hundred)

Using
 $K = 0.1116$
gives
 $P = 61052.38...$
and
 61100 is
population

$$= 7$$

(iv) $t \rightarrow \infty$

$\therefore$ limiting velocity is 1 m s^{-1} ✓ (accept!)

alternate method
and mark allocation.

OR

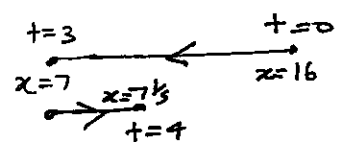
$$t=0$$
$$x=16$$

$$t=3$$
$$x=7$$

$$x = 4 + \frac{16}{9+1}$$

$$= 7\frac{1}{5}$$

$\therefore$ Total distance travelled $= 9 + \frac{1}{5}$
 $= 9\frac{1}{5} \text{ m}$ (✓)



}(✓)

QUESTION 8

$$(a) \frac{dy}{dx} = \frac{2x}{x^2+e}$$

$$y = \int \frac{2x}{x^2+e} dx$$

$$y = \log_e(x^2+e) + C$$

(0, 2)

$$2 = \log_e e + C$$

$$C = 1$$

$$\therefore y = \log_e(x^2+e) + 1$$

$$(b)(i) y = 4 \ln x$$

$$\frac{y}{4} = \log_e x$$

$$x = e^{y/4}$$

$$V = \pi \int_0^4 (e^{y/4})^2 dy$$

$$= \pi \int_0^4 e^{0.5y} dy$$

for 2nd mark
look for squaring of
function and calculation
of 4ln e = 4

$$\begin{cases} x=1 \\ y=0 \\ x=e \\ y=4 \ln e \\ =4 \end{cases}$$

$$(ii) V = \pi \left[\frac{e^{y/2}}{1/2} \right]_0^4$$

$$= \pi(2e^2 - 2 \times 1)$$

$$= 2\pi(e^2 - 1) \text{ units}^3$$

$$(c)(i) \log 3 + \log 9 + \log 27 + \dots$$

$$= \log 3 + 2\log 3 + 3\log 3 + \dots$$

$$T_2 - T_1 = 2\log 3 - \log 3$$

$$= \log 3$$

$$T_3 - T_2 = 3\log 3 - 2\log 3$$

$$= \log 3$$

$$T_3 - T_2 = T_2 - T_1 \therefore \text{A.P.}$$

$$(ii) S_n = \frac{n}{2} [2a + (n-1)d]$$

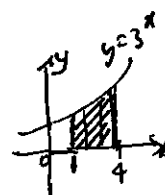
$$S_{10} = \frac{10}{2} [2 \times \log 3 + (10-1) \times \log 3]$$

$$= \frac{10}{2} [2 \log 3 + 9 \log 3]$$

$$= 55 \log 3$$

(d)

x	1	2	3	4
A(n)	3 ¹	3 ²	3 ³	3 ⁴



or equivalent

$$\text{Area} = \frac{1}{2} [3^1 + 2 \times 3^2 + 2 \times 3^3 + 3^4]$$

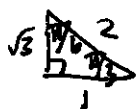
$$= 78 \text{ units}^2$$

QUESTION 9

(a) $\cos x = -\frac{\sqrt{3}}{2}$

$x = \frac{5\pi}{6}, -\frac{5\pi}{6}$

✓✓



(b)(i) $f(x) = e^{\tan x}$

$f'(x) = \sec^2 x e^{\tan x}$

✓

(ii) $\therefore \int 3e^{\tan x} \sec^2 x dx = 3e^{\tan x} + c$

$\int_0^{\pi/4} 3e^{\tan x} \sec^2 x dx = [3e^{\tan x}]_0^{\pi/4}$

✓

$= 3(e^{\tan \pi/4} - e^0)$

$= 3(e - 1)$

✓

(c) (i) 240°

$= 240^\circ \times \frac{\pi}{180}$

$= \frac{4\pi}{3}$

$\left. \begin{array}{l} \text{or } \frac{2}{3} \times 2\pi \\ = \frac{4\pi}{3} \end{array} \right\}$

✓

(ii) $L = r\theta$

$= 12 \times \frac{4\pi}{3}$

$= 16\pi \text{ cm}$

✓

(iii) $a = \frac{1}{2} r^2 \theta$

$= \frac{1}{2} \times 12^2 \times \frac{4\pi}{3}$

$= 96\pi \text{ cm}^2$

✓

(d) (i) $a = 1$

$n = 2$

} need both

(d)(ii) $\text{Area} = \int_{\pi/2}^{3\pi/2} (\sin 2x - 4 \cos x) dx$

✓

$= \left[-\frac{1}{2} \cos 2x - 4 \sin x \right]_{\pi/2}^{3\pi/2}$

✓

$= \left(-\frac{1}{2} \cos 3\pi - 4 \sin 3\pi \right) - \left(-\frac{1}{2} \cos \pi - 4 \sin \pi \right)$

$= \left(-\frac{1}{2} \times -1 - 4 \times -1 \right) - \left(-\frac{1}{2} \times -1 - 4 \times 0 \right)$

$= \left(\frac{1}{2} + 4 \right) - \left(\frac{1}{2} - 0 \right)$

$= 4\frac{1}{2} - \left(-3\frac{1}{2} \right)$

$= 8 \text{ units}^2$

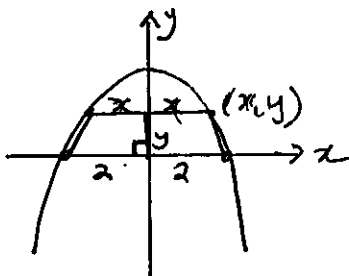
✓

QUESTION 10

(a) $\int_{-2}^5 g(y) dy = 7-9$
 $= -2$ ✓

(b) (i) $\left. \begin{matrix} R(-2,0) \\ S(2,0) \end{matrix} \right\}$ ✓

(ii)



Area $= \frac{1}{2} \times y \times (2x+4)$ ✓ as required
 $= \frac{1}{2} \times (4-x^2) \times (2x+4)$ ✓
 $= (x+2)(4-x^2)$

as required

(iii) $A = 4x - x^3 + 8 - 2x^2$
 $= -x^3 - 2x^2 + 4x + 8$

$\frac{dA}{dx} = -3x^2 - 4x + 4$ ✓

$\frac{dA}{dx} = 0$

$3x^2 + 4x - 4 = 0$

$(3x-2)(x+2) = 0$

$x = \frac{2}{3}$ or -2

$x \neq -2$

$x = \frac{2}{3}$ ✓

$\frac{d^2A}{dx^2} = -6x - 4$

$x = \frac{2}{3}$

$\frac{d^2A}{dx^2} = -6 \times \frac{2}{3} - 4$

$= -8$

< 0 ✓ $\therefore$ maximum value of A when $x = \frac{2}{3}$

x	$\frac{1}{3}$	$\frac{2}{3}$	1
$\frac{dA}{dx}$	$\frac{2}{3}$	0	-3
Sign	+	-	-

(c) $B_1 = 50\,000 \times 1.05 - 1500$ ✓

$B_2 = (50\,000 \times 1.05 - 1500) \times 1.05$
 $- 1500 \times 1.03$ ✓

$= 50\,000 \times 1.05^2 - 1500(1.05 + 1.03)$

$B_3 = [50\,000 \times 1.05^2 - 1500(1.05 + 1.03)] \times 1.05$
 $- 1500(1.03)^2$

$= 50\,000 \times 1.05^3 - 1500(1.05^2 + 1.03 \times 1.05 + 1.03^2)$

Note B_2 and B_3

Marks awarded for indicating the method on previous

"B"

and not just the answer (especially B_3 as it is given in the question!)

(ii) Following this pattern

$B_n = 50\,000 \times 1.05^n$
 $- 1500(1.05^{n-1} + 1.03 \times 1.05^{n-2} + 1.03^2 \times 1.05^{n-3} + \dots + 1.03^{n-1})$

G.P. $a = 1.05^{n-1}$

$r = \frac{1.03}{1.05}$

'n' terms

Using $S_n = \frac{a(r^n - 1)}{r - 1}$

naive

showing how!!

$B_n = 50\,000 \times 1.05^n - 1500 \left[\frac{1.05^{n-1} \left(\left(\frac{1.03}{1.05} \right)^n - 1 \right)}{\frac{1.03}{1.05} - 1} \right]$

$= 50\,000 \times 1.05^n - 1500 \left[\frac{1.05^{n-1} \left(\left(\frac{1.03}{1.05} \right)^n - 1 \right)}{\frac{-0.02}{1.05}} \right]$

$= 50\,000 \times 1.05^n - \frac{1500}{0.02} \left[1.05^n \left(1 - \left(\frac{1.03}{1.05} \right)^n \right) \right]$

$= 50\,000 \times 1.05^n - 75\,000 \times 1.05^n \left(1 - \left(\frac{1.03}{1.05} \right)^n \right)$

given in question AS REQUIRED!